

Lab No. 10

Counters with Non-Binary Sequence

1. Design a counter with following repeated binary sequence: 0, 1, 2, 3, 4, 5, 6, 7. Use JK flip-flops

Excitation Table

Present state			Next state			Flipflop					
A	B	C	A	B	C	J _A	K _A	J _B	K _B	J _C	K _C
0	0	1	0	0	1	0	X	0	X	1	X
0	1	0	0	1	0	0	X	1	X	X	1
0	1	1	1	0	0	1	X	X	1	0	X
1	0	0	1	0	1	X	0	0	X	1	X
1	0	1	1	1	0	X	0	1	X	X	1
1	1	0	1	1	1	X	0	X	0	1	X
1	1	1	0	0	0	X	1	X	1	X	1

AB
C 00 01 11 10

0	0	1	X	X
1	0	X	X	X

$$J_A = B$$

AB
C 00 01 11 10

0	X	X	0	0
1	X	X	1	0

$$K_A = BC$$

AB
C 00 01 11 10

0	0	X	X	0
1	1	X	X	1

$$J_B = C$$

AB
C 00 01 11 10

0	X	1	0	X
1	X	X	1	X

$$K_B = C + \bar{A}$$

c \ AB	00	01	10	11
0	1	0	1	1
1	X	X	X	X

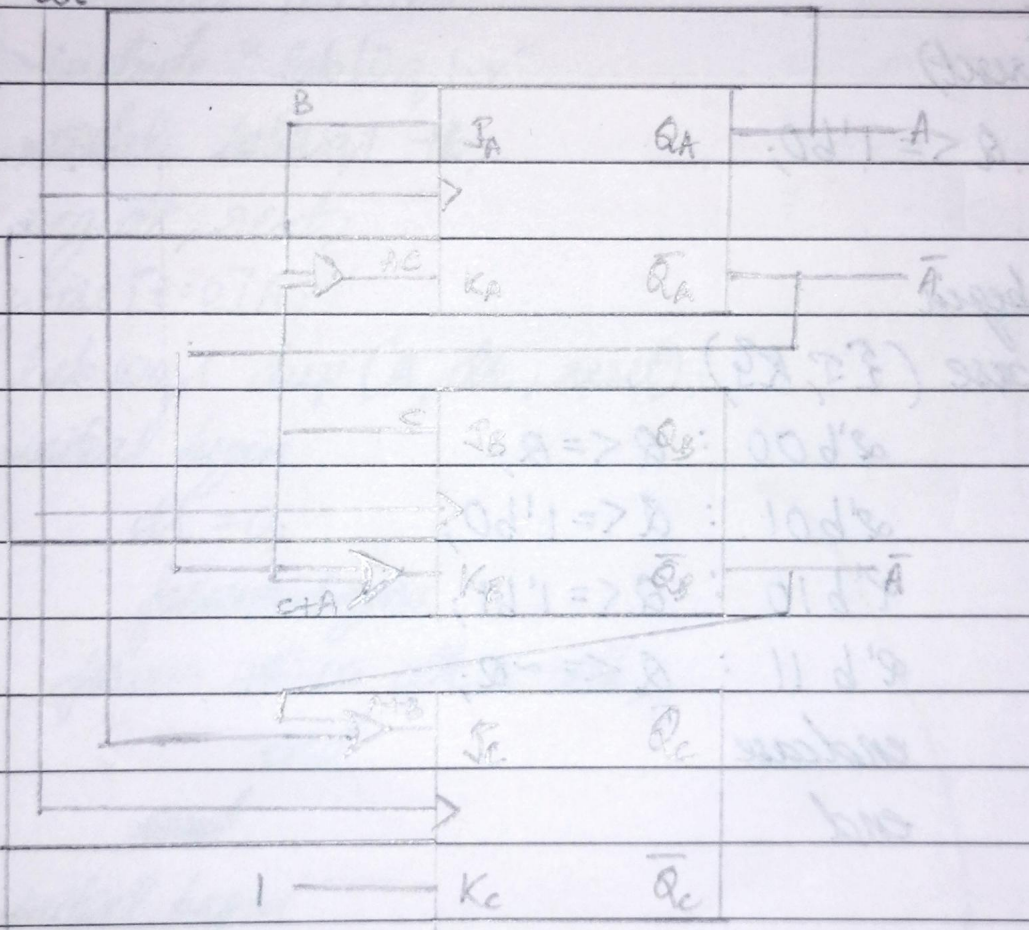
$$I_c = A + \bar{B}$$

c \ AB	00	01	11	10
0	X	X	X	X
1	1	X	1	1

$$K_c = 1$$

Diagram

clk



Reset

code

```

module JKFF (J, K, Q, clk, reset);
input J, K, clk, reset;
output reg Q;
always @ (posedge clk or posedge reset)
begin
    if (reset)
        Q <= 1'b0;
    else
        begin
            case ({J, K})
                2'b00 : Q <= Q;
                2'b01 : Q <= 1'b0;
                2'b10 : Q <= 1'b1;
                2'b11 : Q <= ~Q;
            endcase
        end
end
endmodule

```

```

module tab10q1 (A, clk, reset);
input clk, reset;
output [2:0] A;
JKFF s0 (A[1], A[1] & A[0], A[2], clk, reset);

```

```

J1FF S1(A[0], A[0]~A[2], A[1], clk, reset);
J1FF S2(A[2], ~A[1], 1'b1, A[0], clk, reset);
endmodule

```

Testbench

```

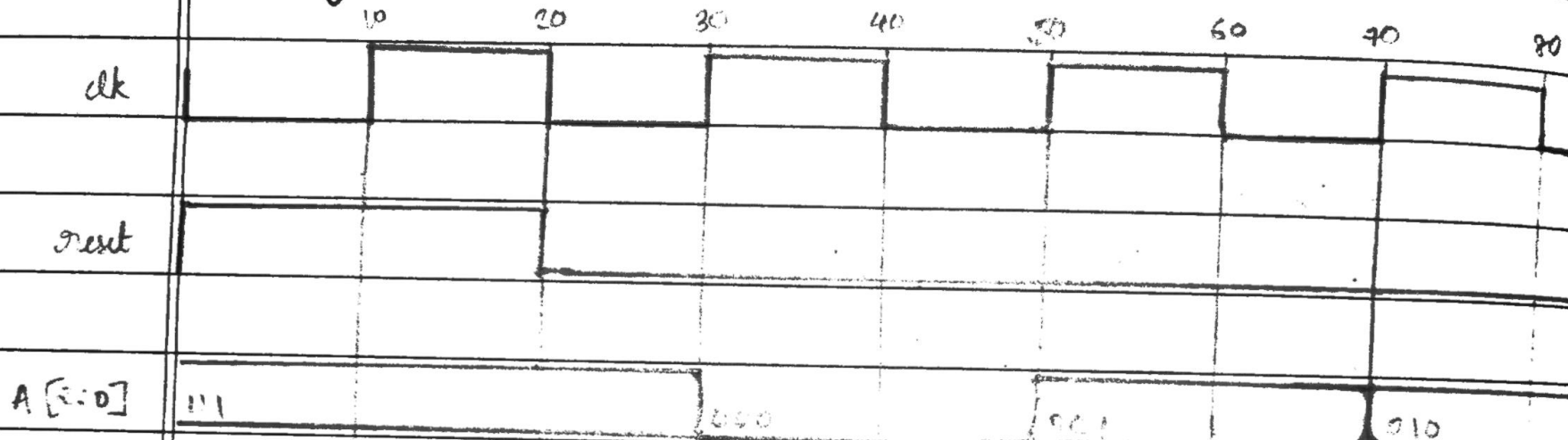
timescale 1ns/1ns
include "lab10q1.v"
module lab10q1_tb;
reg clk, reset;
wire [7:0] A;
lab10q1 dup(A, clk, reset);
initial begin
    clk = 0;
    forever begin
        forever #10 clk = ~clk;
    end
end
end

initial begin
    $dumpfile ("lab10q1_tb.vcd");
    $dumpvars (0, lab10q1_tb);
    reset = 1; #20;
    reset = 0; #800;
    $display("test complete");
    $finish;
end

```

end
endmodule

Waveform



2. Derive the expressions for the design of a counter with the following repeated binary sequence: 1, 2, 3, 0, 7, 6, 5. Use SR flipflops.

Excitation Table

Present State			Next State			Flipflop					
A	B	C	A	B	C	S_A	R_A	S_B	R_B	S_C	R_C
0	0	1	0	1	0	0	X	1	0	0	1
0	1	0	0	1	1	0	X	X	0	1	0
0	1	1	0	0	0	0	X	0	1	0	1
1	0	0	1	1	1	1	0	1	0	1	0
1	0	1	1	1	0	X	0	X	0	0	1
1	1	0	1	0	1	X	0	0	1	1	0
1	1	1	0	0	1	0	1	0	X	X	0

AB	00	01	11	10
0	1	0	X	1
1	0	0	X	0

$S_A = \overline{B}\overline{C}$

AB	00	01	11	10
0	0	X	0	X
1	X	X	0	1

$R_A = \overline{A}\overline{B}$

AB	00	01	11	10
0	1	X	0	X
1	1	0	X	0

$S_B = \overline{A}\overline{B}$

AB	00	01	11	10
0	0	0	1	X
1	0	1	0	X

$R_B = \overline{A}\overline{C} + \overline{A}BC$

AB	00	01	11	10
0	1	1	1	X
1	0	0	0	X

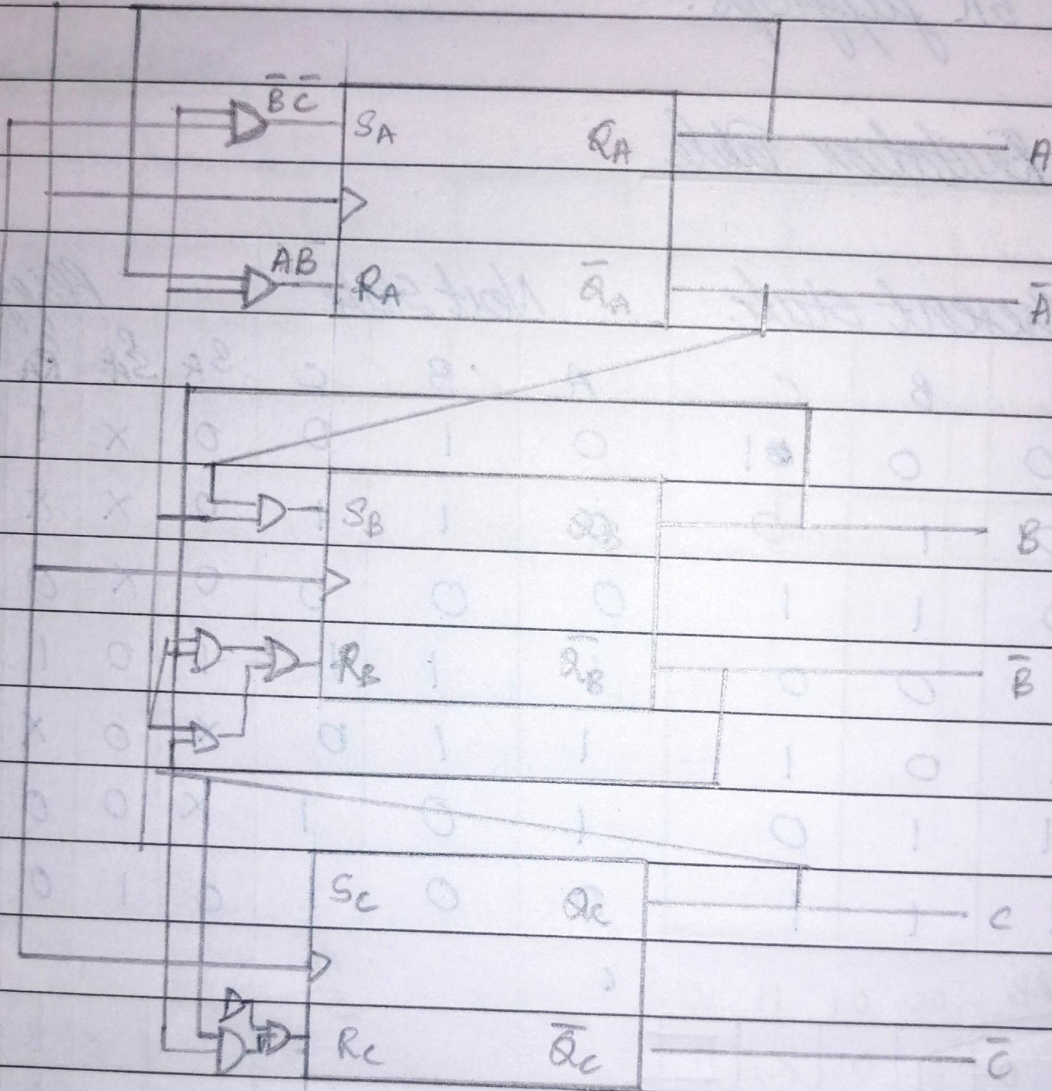
$S_C = \overline{C}$

AB	00	01	11	10
0	0	0	0	X
1	1	1	1	0

$R_C = \overline{A}\overline{C} + BC$

Diagram

clk



Reset

code

```
module SRFF(S, R, Q, clk, reset, inp);  
input S, R, clk, reset, inp;  
output reg Q;  
always @ (posedge clk or posedge reset),  
begin  
    if(reset)  
        Q <= inp;  
    else  
        begin  
            case ({S, R})  
                2'b00 : Q <= Q;  
                2'b01 : Q <= 1'b0;  
                2'b10 : Q <= 1'b1;  
                2'b11 : Q <= 1'bx;  
            endcase  
        end  
    end  
end  
end  
endmodule
```

```
module Lab10q2(A, clk, reset);  
input clk, reset;  
output [2:0] A;  
SRFF S0((-A[1] & ~A[0]), (A[2] & ~A[1]), A[2], clk, reset, 1'b0);
```

```

SRFF s1((~A[2] & ~A[1]), (A[2] & ~A[0]) (~A[2] & A[1] & A[0]), A[1],
        clk, reset, 1'b0);
SRFF s2((~A[0]), ((~A[2] & A[0]) | (A[1] & A[0])), A[0], clk, reset, 1'b0);
endmodule

```

Testbench

```

~timescale 1ns/1ns
~include "lab10q2.v",
module labq2_tb;
reg clk, reset;
wire [2:0] A;
lab10q2 dut (A, clk, reset);
initial begin
    clk = 0;
    forever begin
        #10 clk = ~clk;
    end
end

initial begin
    $dumpfile ("lab10q2_tb.vcd");
    $dumpvars (0, lab10q2_tb);
    reset = 1; #20;
    reset = 0; #100;
    $display ("test complete");

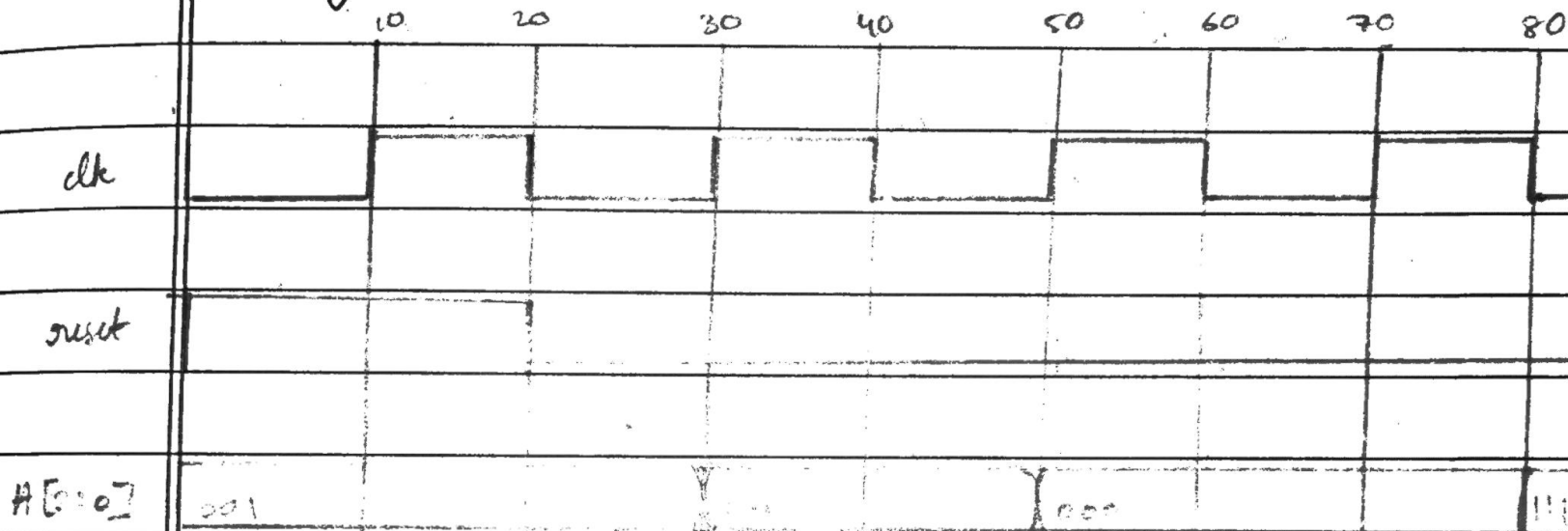
```

PAGE NO :

DATE : 4/11/23

* finish;
end
endmodule

Waveform



3. Design a counter with T-flipflops that goes through the following binary repeated sequence: 0, 1, 3, 7, 6, 4.

Excitation Table

Present State			Next State			Flipflop		
A	B	C	A	B	C	T _A	T _B	T _C
0	0	0	0	0	1	0	0	1
0	0	1	0	1	1	0	1	0
0	1	1	1	1	1	1	0	0
1	1	1	1	1	0	0	0	1
1	1	0	1	0	0	0	1	0
1	0	0	0	0	0	1	0	0

AB

C \	00	01	11	10
0			1	X
1	1	X		

$(A \oplus B)$
 $T_A = A\bar{B} + B\bar{A}$

AB

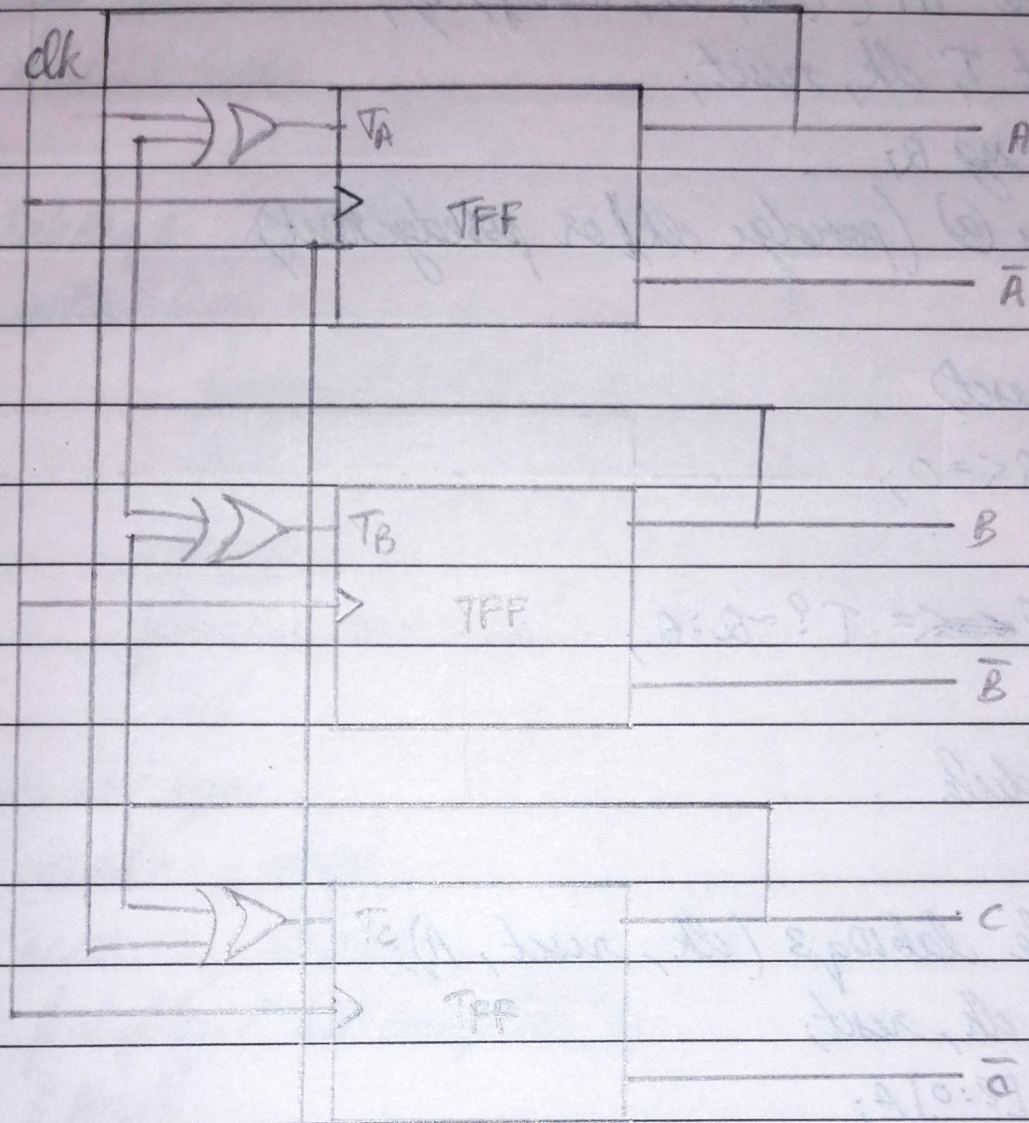
C \	00	01	11	10
0		1		X
1		X		1

$(B \oplus C)$
 $T_B = \bar{B}C + B\bar{C}$

AB

C \	00	01	11	10
0	1			X
1		X	1	

$(A \oplus C)$
 $T_C = AC + \bar{A}\bar{C}$

Diagram

code

```

module TFF (T, clk, clk, reset, A);
input T, clk, reset;
output reg A;
always @ (posedge clk or posedge reset)
begin
    if (reset)
        A <= 0;
    else
        A <= T ? ~A : A;
    end
endmodule

```

```

module lab10q3 (clk, reset, A);
input clk, reset;
output [2:0] A;
TFF s1(A[2] ^ A[1], clk, reset, A[2]);
TFF s2(A[1] ^ A[0], clk, reset, A[1]);
TFF s3(~(A[2] ^ A[0]), clk, reset, A[0]);
endmodule

```

Testbench

```

- timescale 1ns / 1ns
- include "lab10q3.v"

```

```
module lab10q3_tb;  
  reg clk, reset;  
  wire [7:0] A;
```

```
  lab10q3_dut (clk, reset, A);  
  initial begin
```

```
    $dumpfile ("lab10q3_tb.vcd");
```

```
    $dumpvars (0, lab10q3_dut);
```

```
    clk = 0;
```

```
    forever #10 clk = ~clk;
```

```
  end
```

```
  initial begin
```

```
    reset = 1; #20;
```

```
    reset = 0; #280;
```

```
    $display ("test complete");
```

```
    $finish;
```

```
  end
```

```
endmodule
```

waveform

10

20

30

40

50

60

70

80

clk

reset

A[2:0]

000

001

011

111